



Diagnosing the cold start failure of fuel cell system: End-plate cell's thermo-hydraulic imbalance and its mitigation through loading control

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ABSTRACT

Reliable auxiliary heating-free self-startup under low-temperature conditions is crucial for enhancing the environmental adaptability, reducing costs, and expanding the commercial application prospects of fuel cells. However, current research primarily focuses on single-cell levels, while studies on commercial high-power stacks, particularly regarding the influence of startup current loading strategies on the failure mechanisms of end-plate cells, remain insufficient. To address this, this study conducted multiple cold-start experiments under various ambient temperatures and combined them with one-dimensional model simulations to systematically analyze the effects of loading strategies on coolant temperature, stack voltage uniformity, and the performance of end-plate cells. Experimental and simulation results reveal that cold-start failure is primarily caused by the sudden voltage drop of end-plate cells due to the “end-plate effect” and ice blockage in the later stage. The study demonstrates that, compared to the step loading strategy, the linear loading strategy better balances the competition between heat generation and ice formation during startup, allowing system heat generation to gradually dominate. This assists end-plate cells in achieving temperature breakthrough and completing the startup process. Experimental results show that the linear loading strategy significantly improves startup performance, reducing startup time by 7 s and 2 s at $-15\text{ }^{\circ}\text{C}$ and $-20\text{ }^{\circ}\text{C}$, respectively, and successfully extending the auxiliary heating-free startup limit to $-30\text{ }^{\circ}\text{C}$. Furthermore, this strategy effectively enhances stack voltage uniformity, with the voltage non-uniformity coefficient reduced by up to approximately 80.67%. This study provides an effective control method to enhance the all-climate operational capability of PEMFCs.

1. Introduction

The proton exchange membrane fuel cell (PEMFC), a pivotal form of hydrogen energy utilization, has emerged as a strong contender for future transportation powertrains due to its advantages such as high power density, low operational noise, low operating temperature, and pollution-free byproducts [1–3]. It also represents a vital energy utilization pathway for achieving United Nations Sustainable Development Goal 7 (SDG-7). However, achieving true commercial application still faces challenges in durability, cold start capability, and cost. In particular, the difficulty of cold start under frigid conditions severely limits the promotion of fuel cells in all-climate and wide-temperature-range scenarios [4].

At sub-zero temperatures, the freezing and thawing of water within the fuel cell can induce cyclic stresses that damage materials and mechanical structures, while also blocking reactant flow channels and covering active catalyst sites, leading to a sudden performance drop [5]. It is widely recognized that the key to a successful cold start lies in maintaining a dynamic balance between heat generation (which raises the stack temperature) and water production (which risks ice formation) [6–8]. The process is considered successful if the cell temperature rises above the freezing point before the active reaction area becomes completely blocked by ice [8]. Moreover, improper cold start processes can result in unacceptably low fuel cell performance and further impair durability. As a result, effective cold start technology is regarded as a critical bottleneck that must be resolved in this field.

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In recent years, research on cold start has primarily focused on the influence of key operating parameters—such as startup conditions [9–11], membrane electrode assembly (MEA) materials [12], flow field plate design and material [13,14], and heating methods [15–17]—on startup performance. Advanced techniques, including in-situ segmented measurement [18] and X-ray imaging [19], have been employed to observe the distribution characteristics of water, heat, and current inside the cell. Studies indicate that at lower temperatures, ice formation in fuel cells primarily occurs within the cathode catalyst layer; whereas near freezing temperatures, ice tends to form at the interface between the cathode catalyst layer and the gas diffusion layer, becoming the main cause of performance degradation [20,21]. Neutron imaging reveals that the freezing process within the fuel cell initiates in the cathode outlet region and propagates upstream along the cathode gas flow direction, eventually leading to complete freezing in the cathode inlet area, which results in cold start failure [22]. Further insights from in-situ segmented measurement techniques reveal that current density and temperature distributions exhibit dynamic migration characteristics during cold start. For instance, under co-flow configuration with a single-serpentine flow field, the high-current-density region shifts gradually from the inlet toward the middle, accompanied by a corresponding migration of the temperature profile. Reducing voltage load or increasing the initial temperature can significantly shorten the startup time [23,24].

At the stack level, studies emphasize that design parameters such as proton exchange membrane thickness and flow field structure, combined with the pre-start state, collectively determine whether successful startup can be achieved [25]. These factors jointly constrain a minimum startup temperature. Large-scale stacks can achieve ice-free startup through self-start strategies at -20°C ; however, below -30°C , end cells are prone to staged ice formation due to rapid freezing of supercooled water [26].

Therefore, the cold start of PEMFCs is a complex process involving multiple coupled factors, with operational characteristics changing systematically as temperature decreases. Although existing diagnostic techniques can partially capture ice formation and distribution, their application remains challenging at the system level. Thus, there is an urgent need to develop theories and methods for determining the balanced boundaries of water and thermal states applicable to fuel cell systems.

To improve the success rate of cold starts and reduce damage to the internal structure of the fuel cell, a purge strategy is typically employed after shutdown to remove residual water and increase the potential buffer space for ice formation [5,27,28]. However, the cold start process is inherently a race between “ice” and “heat”, whose outcome depends on controlling the rate of temperature increase and the dynamic water-thermal behavior during startup [21,29]. Therefore, gaining a deeper understanding of water/heat transport and phase change mechanisms, and accordingly developing well-matched startup strategies, is crucial to enhancing startup success rates [30].

During low-temperature startup, heat generation is concentrated primarily in the cathode catalyst layer, leading to localized temperature peaks and reduced consistency in temperature and ice distribution among individual cells within the stack [31]. The middle cells, benefiting from smaller heat dissipation area and higher heat utilization efficiency, experience faster temperature rise and demonstrate better performance than those near the end plates, playing a more significant role in the startup process. In contrast, cells adjacent to the end plates remain at lower temperatures with slightly higher ice content, and temperature non-uniformity is further exacerbated as startup progresses [32,33]. Thermal insulation of the end plates has been proven to be an effective measure for improving the heating rate and overall startup performance [34].

Furthermore, notable disparities in current density distribution are observed both among single cells and within each cell. Cells close to the air inlet generally exhibit more uniform gas supply and current

distribution, whereas the central region often shows the highest current density, driving the temperature increase—though local gas starvation may occasionally induce reverse current [35]. Experimental results indicate that at -20°C startup conditions, the voltages of cells at both ends can be significantly lower than the average, a phenomenon described as “low voltage at both ends”, which severely limits startup success and loading rates [36,37]. Repeated cold start cycles can also lead to performance degradation across the stack, with varying degrees of decay among cells. For instance, cells near the end plates often undergo the most pronounced performance loss, and voltage consistency metrics may deteriorate noticeably with cycle number. This inconsistency is mainly attributed to declining water management capability, uneven ice coverage, imbalanced reactant distribution, and significant temperature gradients, collectively causing continuous voltage fluctuations in individual cells during startup [38].

In summary, the development of reliable low-temperature self-startup strategies constitutes a critical bottleneck for achieving the “all-climate, wide-temperature-range” application of fuel cells. The core challenge lies in the dynamic competition between heat generation/temperature rise and water production/ice formation within the cell during startup. While existing research, aided by advanced characterization techniques, has revealed water/ice distribution, temperature, and current density migration patterns at the single-cell or local scale (e.g., ice propagation from the cathode outlet upstream) and identified key influencing factors (e.g., materials, flow field design, operational strategies), significant systemic knowledge gaps persist. The primary issue is that studies predominantly focus on observing and correlating localized phenomena, making it difficult to mechanistically predict and explain the dynamic behavior at the stack/system level. A particularly prominent gap is the lack of in-depth understanding regarding the dynamic evolution patterns and dominant mechanisms behind the severe consistency degradation among individual cells within a stack during cold start—manifested in temperature, voltage, and ice content disparities (such as the initiation and amplification of phenomena like lower voltage in end-plate cells). Therefore, conducting system-level analysis of the cold-start process to establish a theoretical framework capable of quantitatively defining the safety boundaries for system-level water-thermal states, and concurrently developing optimized control strategies that mitigate internal water-thermal competition imbalance and enhance stack consistency, is of paramount importance for improving the cold-start capability and durability of fuel cells.

Given this context, the present study conducts cold start experiments on a fuel cell system under various ambient temperatures, with the goal of improving voltage uniformity among individual cells in the stack. Using a combined experimental and simulation approach, we systematically investigate cold start strategies for fuel cells. The research focuses on the competition between water and heat within the end cells during startup, analyzes and identifies characteristic failure modes of unsuccessful cold starts, and examines the standard deviation of single-cell voltages (C_v) throughout the process. Based on this analysis, the balancing behavior of the stack under different loading strategies is evaluated to identify an optimal loading protocol. The strategy derived from simulations is subsequently applied to cold start tests in a commercial high-power fuel cell system, enhancing the cold start capability of the end cells and thereby improving the overall low-temperature startup performance and reliability of the stack. This work aims to achieve rapid startup under lower temperatures and to provide a theoretical basis for formulating effective cold start strategies.

2. Experimental and theoretical analysis

2.1. Experimental devices

This experiment employed a high-power fuel cell system, placed inside a low-temperature environmental chamber, to conduct a study on its cold start performance at various ambient temperatures. To

thoroughly validate the relevance of the research, the selected fuel cell system's power rating aligns with the practical application requirements of current commercial heavy-duty vehicles.

The schematic diagram of the experimental fuel cell system is shown in Fig. 2 (a), with its relevant parameters provided in Table 1. During the tests, the fuel cell system control unit (FCU) received preset commands from an upper-computer via CAN communication. In accordance with these commands, the FCU utilized its embedded control program to regulate the pressure/flow rate/coolant temperature setpoints for the hydrogen supply system, air supply system, and thermal management system, respectively, thereby meeting the required input parameters for the stack operation. Meanwhile, the stack operating current is controlled by operating the DC/DC converter in a constant current input mode. This integrated control strategy ultimately ensured the stable and normal operation of the entire fuel cell system. To accurately reflect the cold-start capability of the fuel cell system in low-temperature application scenarios, the thermal management system of the fuel cell system (including pipelines, valves, pumps, and radiators) is placed inside the environmental chamber. This ensures adequate heat exchange between the heat generated by the fuel cell and the air inside the chamber during testing. Furthermore, a specialized fuel cell coolant with a freezing point of $\leq -45\text{ }^{\circ}\text{C}$ was used in this test. This was to prevent the coolant from freezing during the experiment and to ensure the test proceeded smoothly. (See Fig. 1.)

The low-temperature environmental chamber used in the experiments is shown in Fig. 2 (b). This equipment is capable of simulating ambient temperature, relative humidity, and high-altitude pressure. The ambient temperature simulation range is from $-40\text{ }^{\circ}\text{C}$ to $+60\text{ }^{\circ}\text{C}$, the relative humidity simulation range is from 20%RH to 90%RH (when the ambient temperature is above $5\text{ }^{\circ}\text{C}$), and the high-altitude pressure simulation range corresponds to elevations from 0 m to 4500 m. For this study, only the ambient temperature simulation function is utilized. The chamber temperature is set to the desired value, and the internal air is cooled accordingly at an average rate of approximately $1.5\text{ }^{\circ}\text{C}/\text{min}$.

2.2. Test protocols

To ensure the accuracy and repeatability of the experimental results, the cold start test in this study follows the procedure shown in Fig. 3. The specific steps are as follows:

(1) Freezing

To ensure that the internal temperature of the stack decreases as much as possible to the simulated ambient temperature and to mitigate the effect of housing insulation on internal cooling, the system is placed at the specified ambient temperature for more than 12 h before testing. After this period, a comprehensive assessment is conducted based on whether the readings from temperature sensors—located near the anode/cathode inlets and the coolant inlet/outlet of the stack—are close to the ambient simulation temperature. This ensures that the internal temperature of the system meets test requirements. If all the above criteria are satisfied, the freezing phase is considered complete, and the system proceeds to the startup stage.

Table 1
Parameters of the fuel cell system.

Parameters	Value	Unit
Rated output power	130	kW
Rated output current	220	A
Rated output voltage	600	V
Number of fuel cells	418	/
Active area	330	cm^2
CVM mode	2 channels per cells	/

(2) Startup

The startup phase is the most critical step in the cold start test. After confirming that both the fuel cell system and the environmental chamber have reached the required temperatures, a startup signal is sent via HMI. Based on data collected from ambient temperature sensors in the air supply system and coolant temperature sensors, the FCU determines whether to activate the cold start mode.

Under this control mode, the fuel cell system performs a cold start using a step-loading strategy. Operating parameters for both the hydrogen and air supply systems are adjusted according to the values specified in Table 2. The cooling system operates in a minor circulation mode with the coolant pump set to 2000 rpm. This speed ensures uniform temperature distribution across the individual cells while minimizing heat loss from the stack, thereby facilitating rapid warm-up.

The current is applied in five steps based on current density: $33\text{ A}@0.1\text{ A cm}^{-2}$, $66\text{ A}@0.2\text{ A cm}^{-2}$, $99\text{ A}@0.3\text{ A cm}^{-2}$, $132\text{ A}@0.4\text{ A cm}^{-2}$, $165\text{ A}@0.5\text{ A cm}^{-2}$. Each current level is maintained for 12 s. The current ramp rate is controlled by the FCU according to the set values provided via the HMI.

After the stack establishes the open circuit voltage (OCV), the fuel cell system begins applying a current starting at 33 A. As the cold start proceeds, the coolant temperature gradually increases. Once the FCU detects that the coolant temperature has exceeded $5\text{ }^{\circ}\text{C}$, the cold start mode is terminated, and the system switches to normal operation mode. Under this mode, the fuel cell system adaptively adjusts its output according to the requested power setpoint to meet the power demand.

To prevent performance degradation of individual cells due to ice formation during cold start, two levels of cell voltage protection thresholds are implemented: 0.2 V and 0.15 V. During startup, if any single cell voltage drops below 0.2 V, the load current is gradually reduced to prevent further voltage decline. Current loading resumes according to the cold start strategy once the voltage recovers above 0.2 V. If any cell voltage falls below 0.15 V, the fuel cell system immediately stops current loading, indicating a failed cold start, and initiates a thawing procedure.

Due to the slow heating rate of the environmental chamber, the thawing process primarily relies on heating the coolant to warm the entire stack internally. The heat source for the coolant comes from two main pathways:

- The compressed air from the air compressor carries thermal energy and transfers heat to the coolant via the intercooler. The compressed air is then directed through the intercooler and a bypass valve into the air exhaust line, preventing any potential impact on the stack's lifespan.
- The circulating coolant pump heats the coolant by pressurizing and performing work on it.

The duration of the thawing process depends on the internal temperature of the stack at the time of failure. Based on experimental observations, once the coolant outlet temperature reaches or exceeds $5\text{ }^{\circ}\text{C}$, the frozen cells inside the stack are generally fully thawed. At this point, the fuel cell system can resume the cold start procedure and proceed to the activation stage.

(3) Activation

To ensure uniform conditions across all cells in the stack—particularly the membrane hydration level—an activation procedure is performed after the cold start phase. Once the coolant outlet temperature reaches or exceeds $5\text{ }^{\circ}\text{C}$, the FCU switches from cold start mode to normal operation mode. It then gradually increases the stack current to the activation operating point ($165\text{ A}@0.5\text{ A cm}^{-2}$) based on the output power request set via the HMI.

After the coolant inlet temperature of the fuel cell system gradually

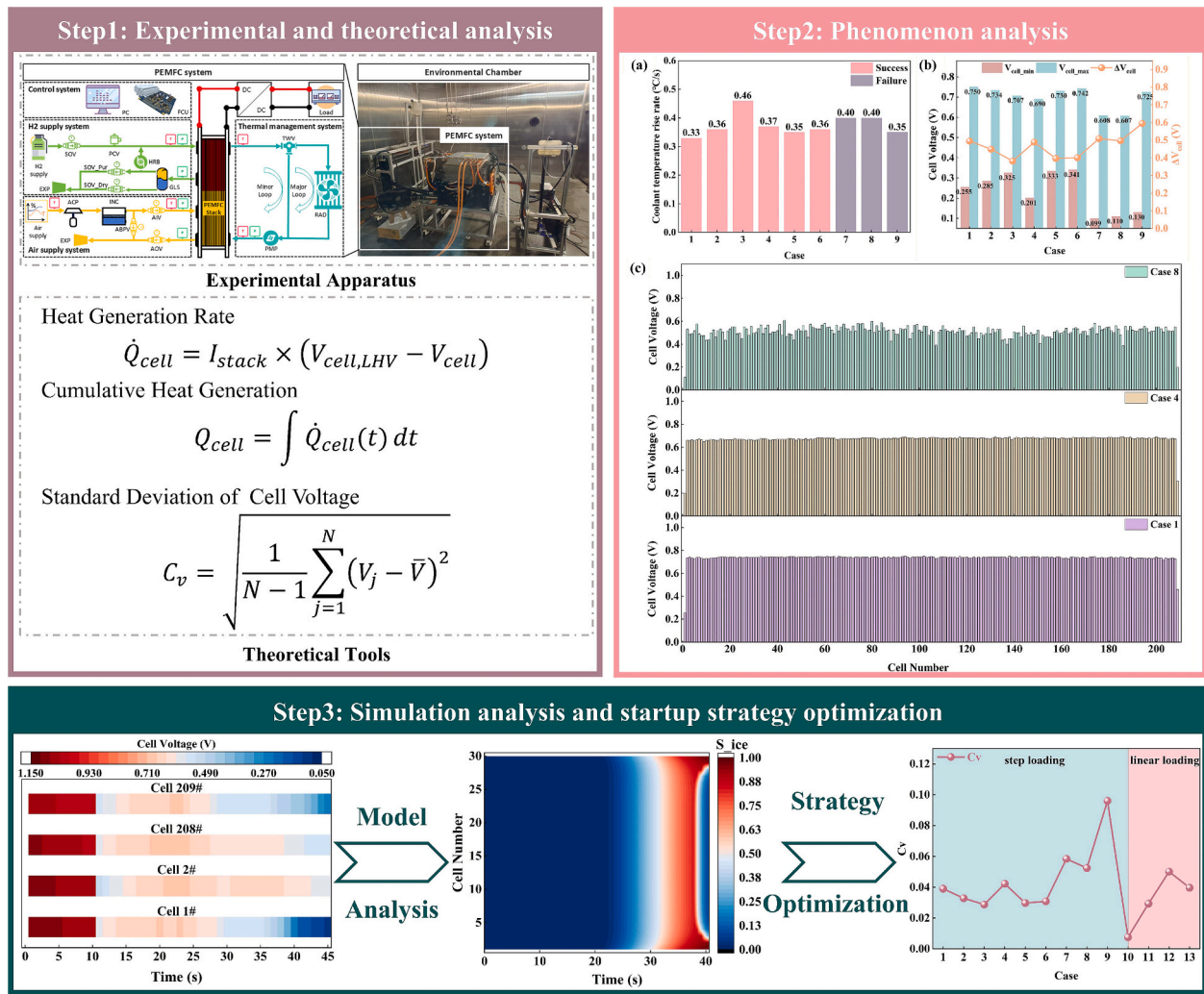


Fig. 1. The basic framework of the paper.

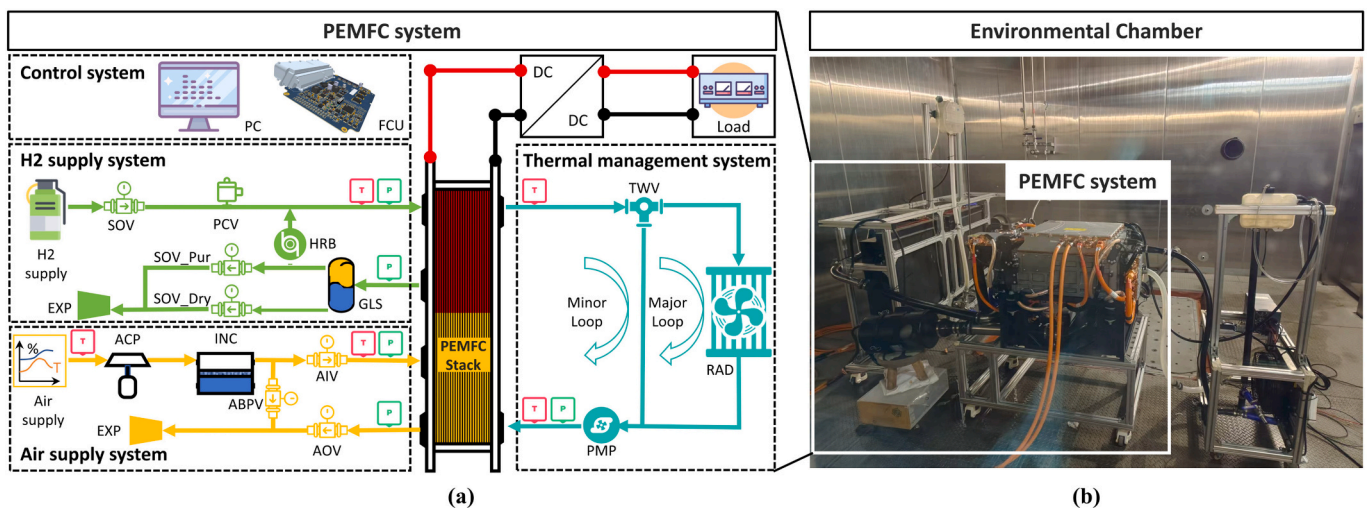


Fig. 2. Schematic of the experimental setup: (a) fuel cell system schematic; (b) low-temperature environmental test chamber.

reaches the required value for activation operation, the system maintains these conditions and operates steadily for 10 min. Upon completion of this stabilization period, the activation process is concluded, and the system proceeds to the purge phase.

(4) Purging

The purpose of the purging process is to reduce the membrane hydration level in each cell of the stack through a series of operational

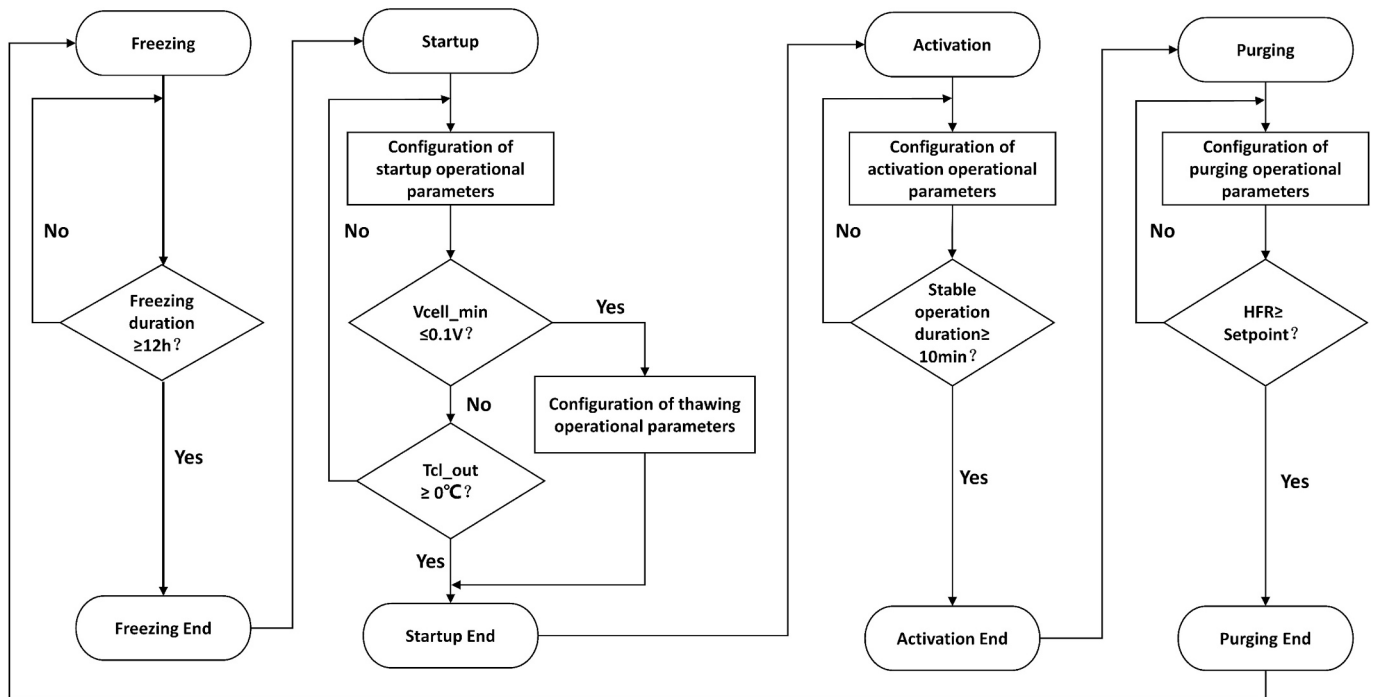


Fig. 3. Flowchart of cold start test for fuel cell system.

Table 2
Operational parameters during cold start of the fuel cell system.

Stack current (A)	Anode inlet pressure (kPa.A)	Cathode stoichiometric ratio (/)	Cathode inlet pressure (kPa.A)	Coolant temperature (°C)
33	142.5	7.0	127.5	50
66	143.0	4.0	128.0	50
99	148.5	2.8	133.5	50
132	160.0	2.5	145.0	55
165	178.5	2.2	163.5	60

steps. This prevents the membrane electrode assembly from freezing due to excessive water content during sub-zero conditions, which could lead to pinhole formation, cracking, or irreversible damage to cell performance.

When a shutdown command is received from the HMI, the fuel cell system initiates the shutdown procedure. The FCU first checks whether the ambient temperature is below 5 °C. If so, the system enters the low-temperature shutdown process. During this phase, a loaded purging mode is used to achieve uniform conditions across the cells. The stack output current is maintained at 40 A, and the coolant inlet temperature is controlled at 50 °C. Hydrogen and air are supplied to purge the stack.

The EIS function integrated into the DC/DC converter (with an excitation current of 5 A and excitation frequency of 1 kHz) is employed to monitor and control the membrane hydration status via the high frequency resistance (HFR) value. The duration of the purging phase is determined by either the HFR reaching a predefined threshold or a maximum allowable purging time. If the target HFR is reached first, the system completes purging and proceeds to shut down. Otherwise, the system continues purging until the maximum time elapses before shutting down.

During shutdown, the DC/DC converter is first disconnected from the stack. Then, the air supply, hydrogen supply, and cooling systems are sequentially deactivated. The system then enters a freezing state, awaiting the next startup.

2.3. Theoretical analysis

(1) Maximum single-cell voltage difference

The maximum single-cell voltage difference, denoted as ΔV is calculated as follows:

$$\Delta V_{\text{cell}} = V_{\text{cell_max}} - V_{\text{cell_min}} \quad (1)$$

Where $V_{\text{cell_max}}$ is the maximum cell voltage value, $V_{\text{cell_min}}$ is the minimum cell voltage value and ΔV_{cell} is the maximum cell voltage difference.

(2) Single-cell heat generation rate

The single-cell heat generation rate is calculated as follows:

$$Q_{\text{cell_dt}} = I_{\text{stack}} \times (V_{\text{cell_LHV}} - V_{\text{cell}}) \quad (2)$$

Where, $Q_{\text{cell_dt}}$ represents the heat generation rate of a single cell, in J/s; $V_{\text{cell_LHV}}$ is the theoretical voltage of hydrogen at the lower heating value, taken as 1.254 V; V_{cell} is the operating voltage of the single cell, in V; I_{stack} denotes the stack current, in A.

(3) Cumulative heat generation of a single cell

The cumulative heat generation of a single cell is calculated as follows:

$$Q_{\text{cell}} = \int \dot{Q}_{\text{cell}}(t) dt \quad (3)$$

Where, Q_{cell} represents the cumulative heat generation of a single cell, in J, $\dot{Q}_{\text{cell}}$ is the heat generation rate of the single cell, in J/s.

(4) Standard deviation of single cell voltages

The standard deviation of single-cell voltages is calculated as follows:

$$C_v = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (V_j - \bar{V})^2} \quad (4)$$

Where, N is the number of single cells, V_j is the voltage of the j^{th} cell, V_{is} is the average cell voltage, C_v is the standard deviation of the cell voltage.

3. Results and discussion

3.1. Experimental phenomenon analysis

In this study, the cold start of the fuel cell was achieved not by preheating the coolant, but by directly applying current to induce internal joule heating for self-startup. This approach is adopted to accurately evaluate the inherent cold-start capability of the fuel cell stack.

To investigate the cold start capability of the fuel cell system under different temperatures, this study conducted a total of nine cold start experiments under three simulated environmental temperatures: -15°C , -20°C , and -25°C . Three repeated trials are performed at each temperature to ensure the reproducibility of the results. Specific experimental parameters and outcomes are summarized in Table 3.

The competition between heat generation and ice formation within the stack is a critical factor determining the outcome of cold start tests. As shown in Table 3, the step-loading strategy enabled successful startups at -15°C and -20°C , with completion times consistently around 50 s. However, all attempts failed at -25°C .

Fig. 4 (a) presents the coolant temperature rise rate during startup from Case 1 to Case 9. At -15°C (Cases 1–3), the temperature rise rate remained above 0.33°C/s , supporting successful initiation. When the ambient temperature decreased to -20°C (Cases 4–6), the rate increased to 0.35°C/s , indicating a greater demand for heat to prevent ice formation within the stack. Despite a further increase to 0.40°C/s at -25°C (Cases 7–9), startup still failed, with the coolant temperature stagnating near -6°C . This suggests that the stack's heat generation rate is insufficient to inhibit the phase transition of membrane-bound water into ice, leading to failure.

Fig. 4 (b) compares single-cell voltages at the end of the startup process. A pronounced disparity between the minimum and maximum cell voltages is observed across cases. In the failed attempts (Cases 7–9), the lowest cell voltage fell below 0.15 V, with a maximum voltage difference ΔV exceeding 0.5 V. Such significant variation indicates highly uneven heat distribution within the stack, resulting in inconsistent startup performance among individual cells.

Fig. 4 (c) visually highlights the performance variations among cells and clearly shows that the lowest voltages occur at both ends of the stack, demonstrating a distinct “end-plate effect.” In successfully started cases (e.g., Case 1 and Case 4), only the edge cells (cell 1# and cell 209#) exhibited slightly lower voltage, while the central cells maintained good voltage uniformity, indicating relatively balanced heat distribution in the middle region where heat generation outpaced ice formation. In contrast, Case 8 not only showed voltage depression at the ends but also a jagged voltage profile in central cells. This implies that the end-plate effect extended toward the center, disrupting heat distribution and leading to divergent outcomes in the competition between

heat generation and ice formation across the stack.

Fig. 5 (a) and (b) show the startup current profile and the corresponding coolant temperature change during Cases 7 to 9, respectively. As the current increased gradually, the coolant temperature remained nearly constant within the first 23 s due to the large thermal mass of the fuel cell stack and the cooling circuit. After 23 s, the coolant temperature began to increase linearly until the cold start ultimately failed.

As illustrated in Fig. 5 (c), the cell voltage distributions at the end of startup in Cases 7–9 exhibited similar characteristics, all showing notably low voltages at both ends and a jagged, non-uniform profile in the middle. Case 9 displayed more pronounced voltage depression across multiple central cells, indicating poorer internal heat distribution compared to Cases 7 and 8. This observation is further supported by the lower coolant temperature recorded in Case 9 at the end of the startup process. The low voltage observed in the cells other than Cell 1# and Cell 209# in Case 9 can be attributed mainly to two factors. First, the relaxation heterogeneity in membrane-bound water redistribution after shutdown and purging led to variations in initial water content among individual cells during the idle period. Cells with higher initial water content entered the “heat generation–ice formation” competition stage earlier during the cold start, resulting in lower output voltage compared to cells that had not yet entered this stage. Second, uneven gas flow distribution within the stack caused asynchronous ice formation across the cells. In some cells, pores in the gas diffusion layer and microporous layer became blocked prematurely due to ice formation, hindering the transport of reactant gases to the reaction zones and leading to local gas starvation, which further reduced the output voltage of these cells.

3.2. Analysis of end-plate cell voltage behavior

Fig. 6 shows the voltage evolution of the end-plate cells (Cell 1# and Cell 209#) and adjacent cells (Cell 2# and Cell 208#) during the cold start in Cases 7–9. The voltage trends are nearly identical across all cases within the first 30 s. Between 30 s and 40 s, Cells 1# and 209# exhibited a more pronounced voltage drop compared to Cells 2# and 208#. After 40 s, the voltages decreased rapidly, with Cell 1# showing the most significant decline. These results indicate that the primary reason for startup failure is the severe performance degradation of the outermost cell (Cell 1#) near the endplate in the later stage, where its voltage fell below the protection threshold.

To further understand the behavior of Cell 1#, Fig. 7 provides a detailed analysis of its performance during startup. As shown in Fig. 7 (a), which compares the voltage profile of Cell 1# under simulated -25°C conditions, the cell remained at OCV during 0–10 s. After current loading began at 10 s, the voltage dropped sharply before gradually recovering. It remained relatively stable above 0.5 V until around 35 s, after which it decreased rapidly until the end of startup. In both Case 7 and Case 9, the voltage of Cell 1# fell below the protection threshold, directly causing startup failure.

Fig. 7 (b) compares the voltage behavior of Cell 1# in Cases 1, 4, and 8 at different temperatures. The voltage trends are consistent with Fig. 7 (a) in the first 35 s. Beyond that point, the voltage dropped sharply in all cases. The rate of voltage decrease is 0.0226 V/s in Cases 1 and 4, and 0.0259 V/s in Case 8.

Fig. 8 compares the startup processes and heat generation characteristics of Cases 1, 4, and 8, dividing the cold start into three characteristic stages based on the applied current profile:

Stage 1: Initialization (0–10 s)

No current is applied during the first 10 s for two main purposes: First, supplying hydrogen/air helps the fuel cell establish OCV, allowing detection of possible cell freezing or membrane puncture. As shown in Fig. 8 (b), the cell voltages all remain above 0.9 V, indicating normal cell conditions. Second, the supplied gas helps purge any residual ice particles from the internal common flow channels or hydrogen/air supply lines, preventing blockages in the anode/cathode flow channels during startup that could lead to reactant starvation and startup failure.

Table 3
Cold start results with step loading.

Case	Ambient temperature ($^\circ\text{C}$)	Current loading rate (A/s)	Operating time (s)	Coolant temperature ($^\circ\text{C}$)	Result
1	-15	4	51	0.10	Success
2	-15	4	52	0.54	Success
3	-15	4	36	0.39	Success
4	-20	4	53	0.19	Success
5	-20	4	53	0.28	Success
6	-20	4	51	0.26	Success
7	-25	4	45	-6.40	Failure
8	-25	4	52	-2.43	Failure
9	-25	4	45	-6.86	Failure

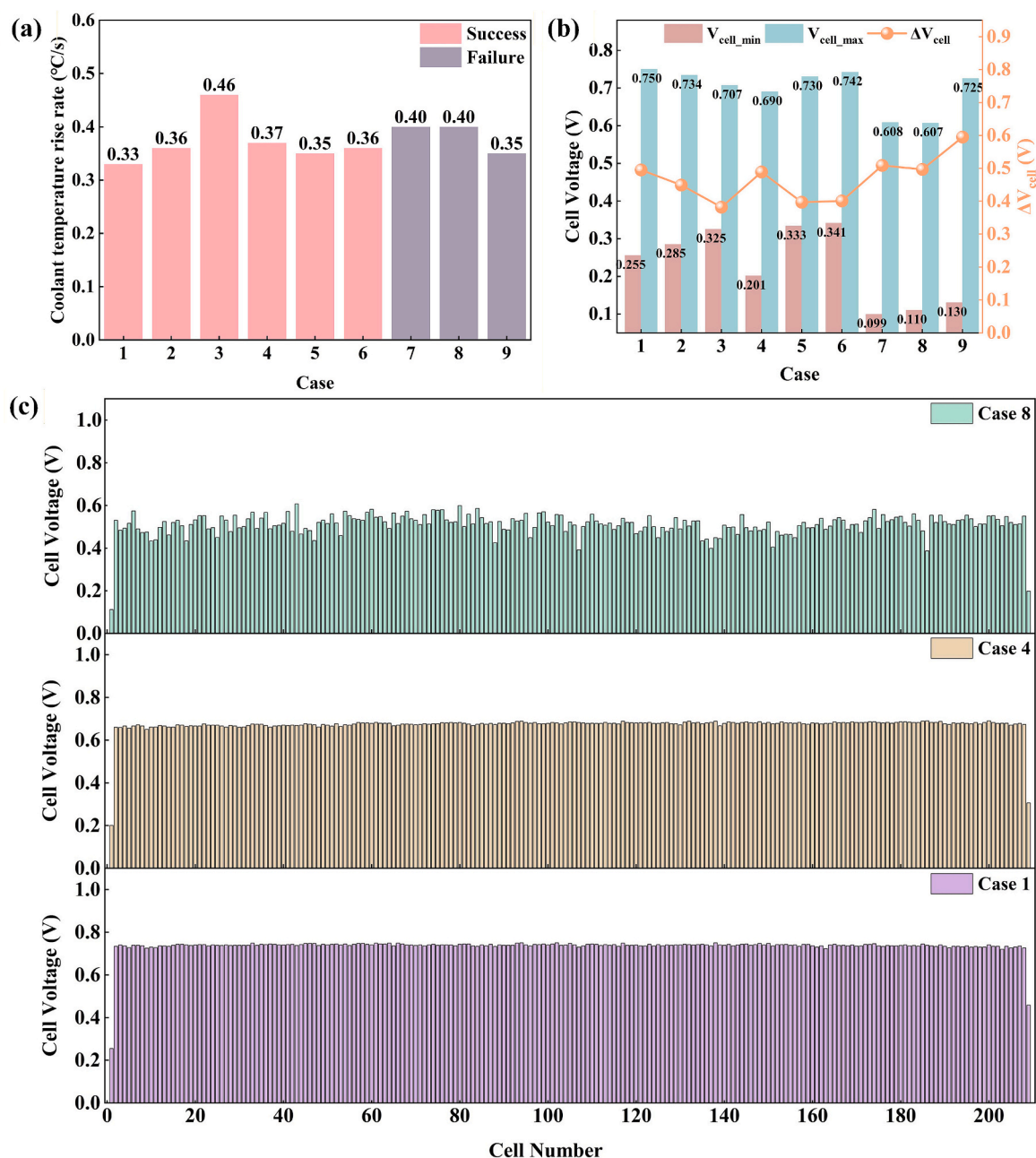


Fig. 4. Cold start process: (a) Coolant temperature rise rate; (b) Cell voltage distribution at the end of startup; (c) Cell voltage profile along the stack at the end of startup.

Stage 2: Membrane Hydration Saturation (10–35 s)

Current loading begins at 10 s, initially set to 33 A (0.1 A cm^{-2}). As shown in Fig. 8 (b), the cell voltage shows an undershoot due to low initial membrane water content, then gradually recovers as hydration increases. Meanwhile, Fig. 8 (c) indicates a slow decrease in the heat generation rate. From 23 s to 35 s, the current is ramped up to 66 A (0.2 A cm^{-2}). The cell voltage experiences a minor drop but recovers once the current stabilizes, suggesting that membrane hydration in Cell 1# has not yet reached saturation.

Stage 3: Competition Phase (35 s onward)

In this stage, liquid water begins to emerge after membrane saturation, initiating competition between heat generation and ice formation. From 35 s to 45 s, the current is increased to 99 A (0.3 A cm^{-2} , slightly delayed in Case 8). As shown in Fig. 8 (b), cell voltages drop by 0.41 V, 0.22 V, and 0.30 V, respectively. Meanwhile, Fig. 8 (c) shows that the

heat generation rate follows Case 8 > Case 1 > Case 4. After 45 s, the current is maintained at 99 A, but the voltage continues to decrease gradually. The heat generation rate shifts to Case 8 > Case 4 > Case 1, indicating that ice has started to form inside the cells and is continuing to accumulate. Ultimately, Cases 1 and 4 complete a successful cold start without triggering the voltage protection threshold once the coolant outlet temperature reaches 0°C , while Case 8 fails.

According to the analysis of cumulative heat generation in Fig. 8 (d), although the total heat generated in Case 1 is slightly lower than that in Cases 4 and 8, it still achieved a successful cold start under a lower simulated ambient temperature. In contrast, the overall cumulative heat generation in Case 4 and Case 8 is essentially the same. However, significant differences in voltage performance emerged toward the end of the startup process: the cell voltage in Case 4 remained at 0.221 V, while in Case 8 it dropped to 0.112 V—a difference of approximately 0.109 V.

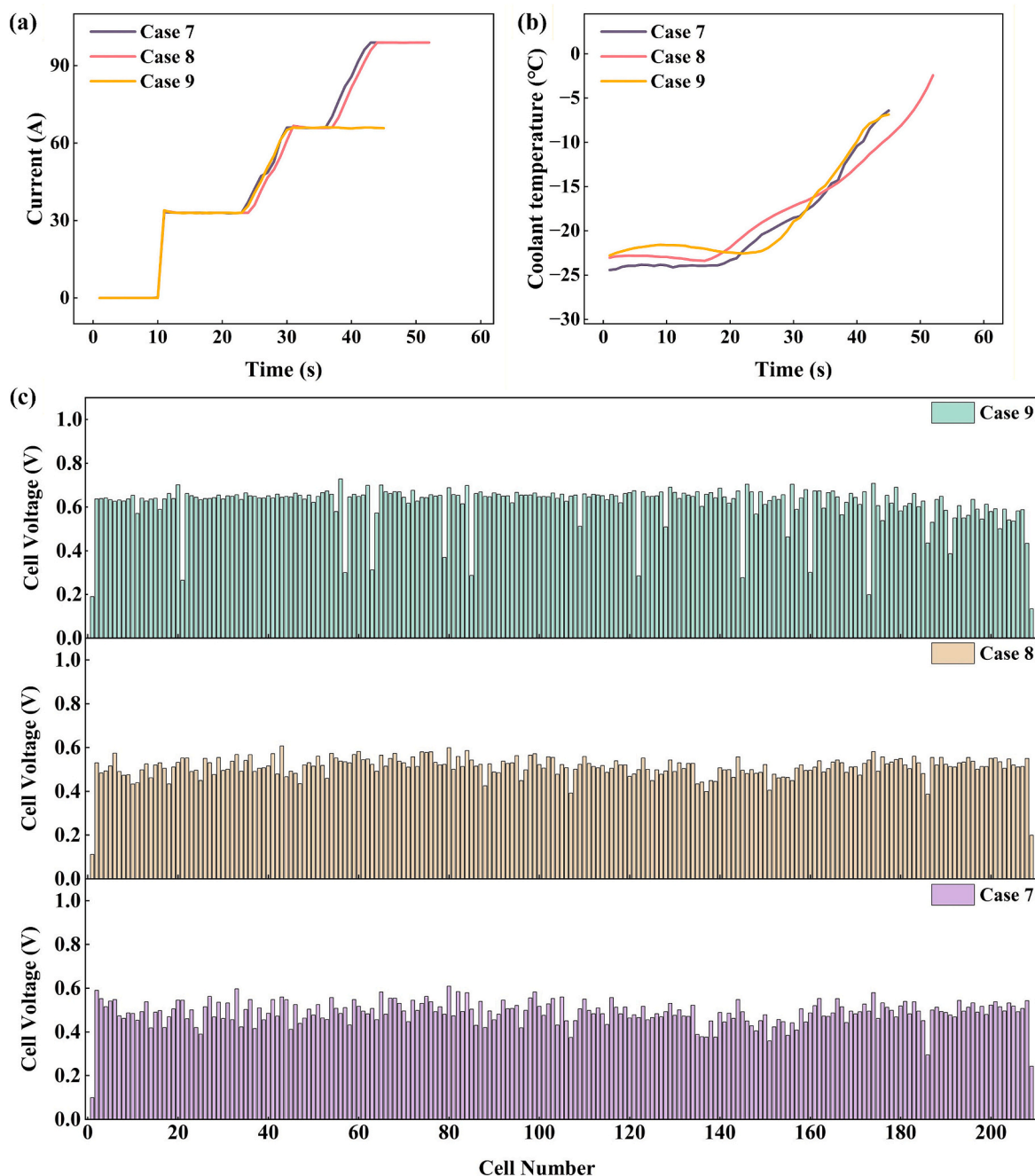


Fig. 5. Cold start processes for Cases 7–9: (a) Applied current; (b) Coolant temperature; (c) Cell voltage distribution at the end of startup.

These results indicate that, during the final stage of cold start, despite similar total heat generation, Case 8 may have suffered from insufficient heat distribution or utilization efficiency. As a result, the amount of heat actually available to mitigate ice formation is lower in Case 8 than in Case 4, ultimately failing to prevent continuous ice accumulation within the cell. This led to complete coverage of the reactive area in Cell 1# by ice and resulted in startup failure.

3.3. Model-based analysis of the cold start process of end-plate cells

Building on previous research [29,39,40], this section employs a one-dimensional low-temperature cold start model of a fuel cell stack consisting of 30 single cells to analyze the cold start behavior of end-plate cells, with the aim of gaining deeper insight into the internal state changes of end-plate cells during cold start. The model simulates

the cold start process at -10°C . Except for the applied current strategies—step-loading and linear-loading (Fig. 9a)—all other parameters remain unchanged.

Fig. 9(b) shows the temperature rise profiles during startup for the endplate-side Cell 1# and the centrally located Cell 15# within the stack. It can be observed that the linear loading strategy enables a rapid temperature increase in the cells. Even for Cell 1#, which is susceptible to the “endplate effect,” its temperature trend remains largely consistent with that of the central cell under the step-loading strategy. This indicates that the linear loading strategy provides a more favorable thermal environment for the cells.

Regarding the changes in ice volume fraction (S_{ice}) within the cells (Fig. 9c), ice gradually begins to form in both cells during the early and middle stages of startup. However, as heat generation increases, the centrally located Cell 15# under the linear loading strategy initiates

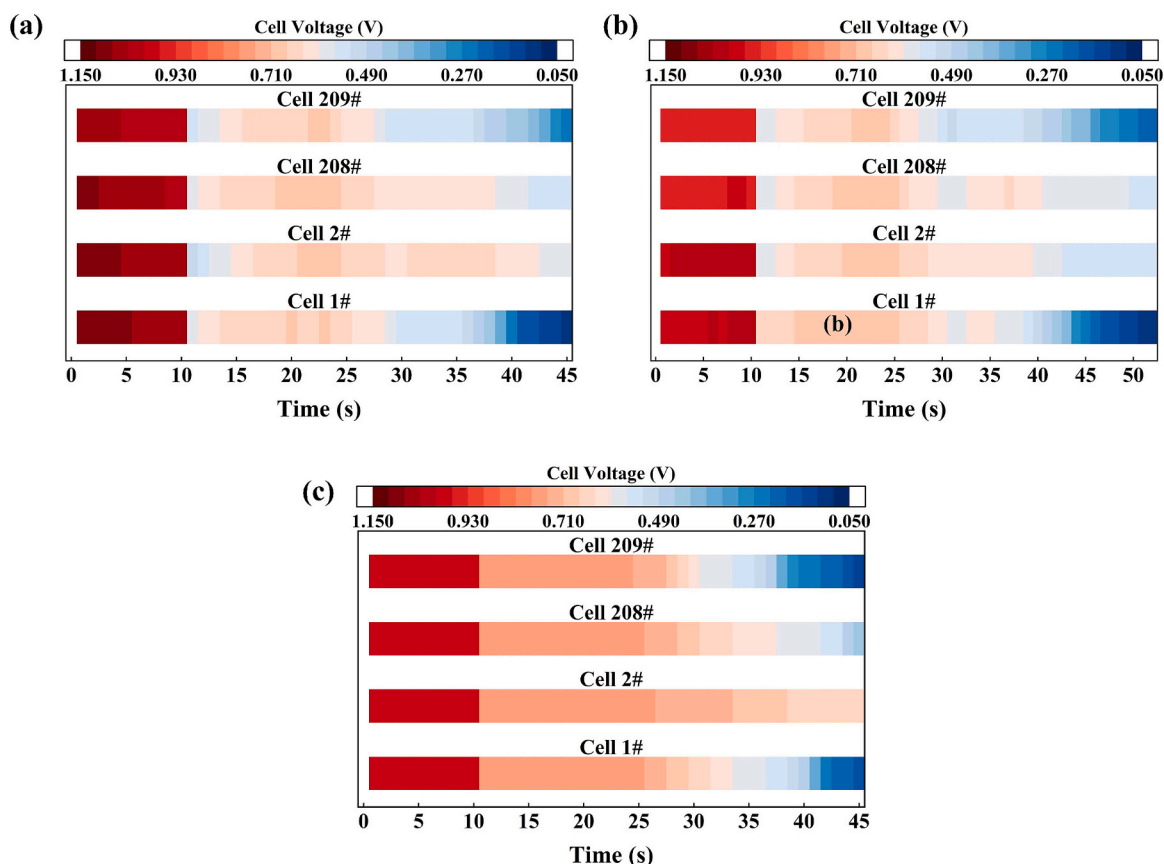


Fig. 6. Voltage evolution of end-plate cells during cold start at $-25\text{ }^{\circ}\text{C}$: (a) Case 7 ; (b) Case 8 ; (c) Case 9.

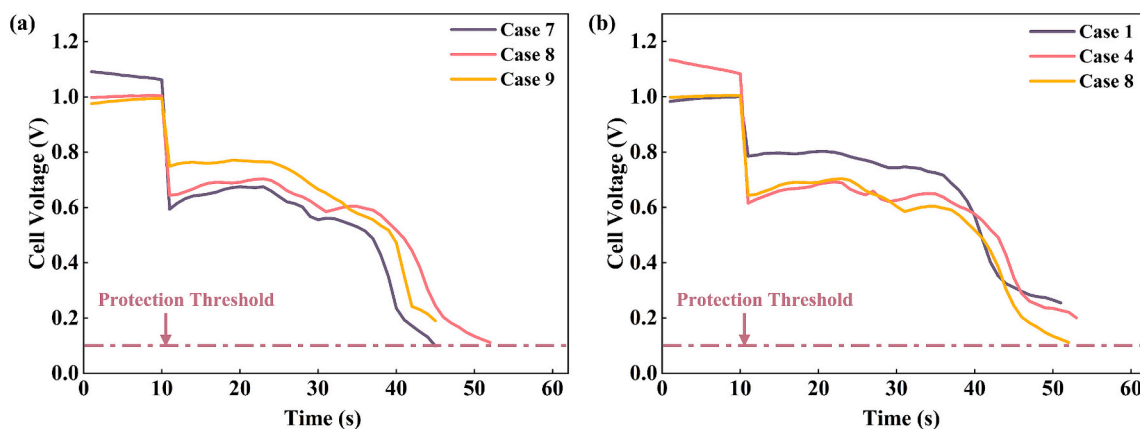


Fig. 7. The voltage of Cell 1# during the cold start process under: (a) Case 7-Case 9; (b) Case 1, Case 4, and Case 8.

melting once the S_{ice} reaches 0.71. The ice content drops sharply and continues to decrease until the end of startup. Although the endplate-side Cell 1# does not begin melting, excess heat from the central cell is circulated to Cell 1# via the coolant flow, raising its temperature and slowing down the ice formation rate. By the end of startup, the S_{ice} in Cell 1# is 0.964, indicating that the reactive zone is not completely ice-covered, resulting in a successful start.

In contrast, under the step-loading strategy, the central Cell 15# begins melting only after the S_{ice} reaches 0.822. A significant portion of the heat is consumed to melt ice in the catalyst layer reaction zone, leaving insufficient residual heat to support the temperature rise of the endplate-side Cell 1#. This ultimately leads to startup failure.

This process is further illustrated by the rate of change in ice volume

fraction (S_{ice_dt}) shown in Fig. 9(d). For Cell 15# under the linear loading strategy, the rate decreased from $+0.051$ to -0.519 , indicating that heat generation overwhelmingly dominated the competition between “heat generation and ice formation.” As a result, the rate of ice accumulation in the endplate-side Cell 1# is also reduced, delaying the complete coverage of active sites in the catalyst layer by solid ice. This extended the time window during which the cell could heat up above $0\text{ }^{\circ}\text{C}$, significantly improving the likelihood of a successful cold start.

As shown in Fig. 9(e) and (f), the linear loading strategy provided a more favorable thermal environment, effectively limiting the expansion of ice volume fraction across the stack—particularly in the endplate-adjacent Cells 1# and 30#. Therefore, compared to the step-loading strategy, the linear loading approach more effectively mitigates

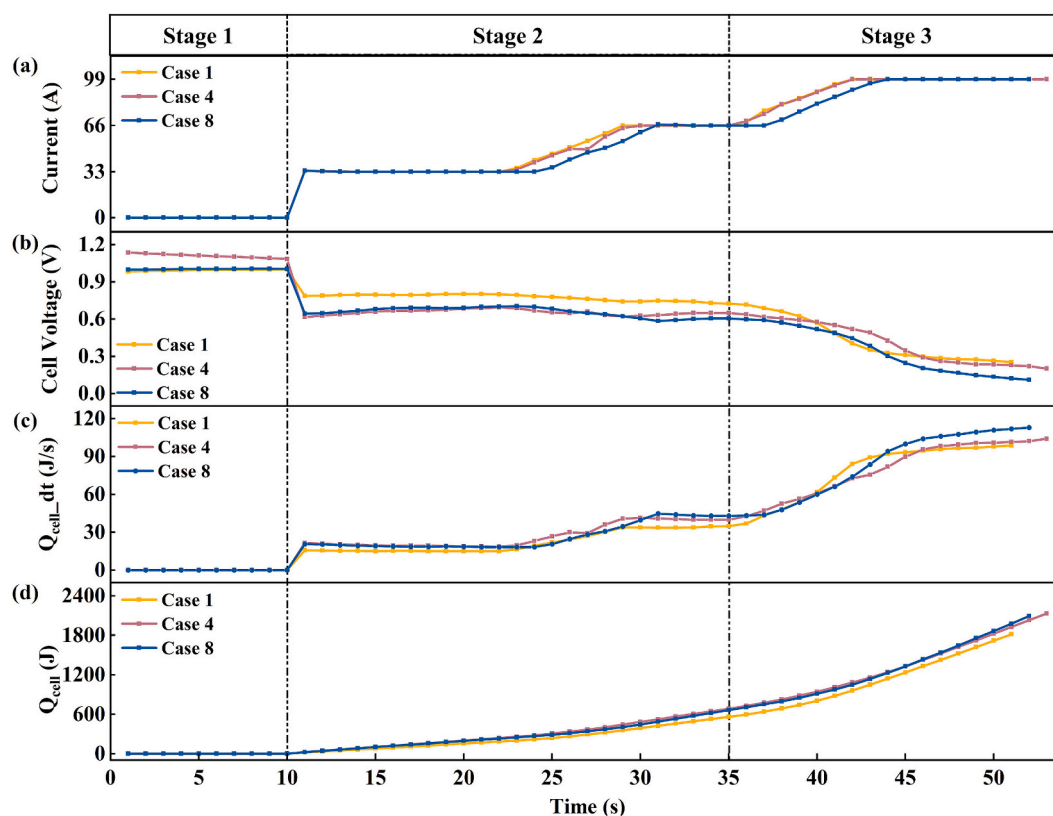


Fig. 8. Cold start process of Cell 1#: (a) Stack current; (b) Cell voltage; (c) Heat generation rate; (d) Cumulative heat generation.

internal ice formation in end-plate cells during cold start, thereby increasing the probability of a successful cold start.

3.4. Influence of step and linear loading strategies on endplate cell voltage

Based on the findings discussed in Section 3.3, this section presents three sets of cold start experiments using a linear loading strategy to investigate its effect on the voltage behavior of end-plate cells. The experimental results are summarized in Table 4.

Fig. 10 presents the experimental results of the cold start under a linear loading strategy. As shown in Fig. 10(a), a small current is applied at the initial startup stage to preheat the stack and improve the uniformity of the initial membrane hydration state across individual cells [28]. Subsequently, the current is increased linearly from 30 A at a specified ramp rate. Although Cases 10 to 12 have similar current ramp rates, resulting in nearly identical coolant temperature profiles (Fig. 10b), the voltages of the end-plate cells (Cell 1# and Cell 209#) still deviated from those of the central cells due to the influence of ambient temperature.

In Fig. 10(c), the end-plate cells in Case 12 began to exhibit generally lower voltages during the mid-startup stage, which differed from Cases 10 and 11 where only Cells 1# and 209# showed voltage depression. Despite the overall voltage decrease in end-plate cells, a successful cold start is still achieved, with the lowest endplate cell voltage remaining at 0.18 V—above the threshold for single-cell undervoltage protection.

These results indicate that at $-25\text{ }^{\circ}\text{C}$ with a linear current ramp rate of 4 A s^{-1} , the “Stage 3: Competition Phase” described in Fig. 8 occurred prematurely. An intense competition between ice formation and heat generation led to voltage depression in the end-plate cells, and the “endplate effect” propagated further into the inner cells of the stack. Therefore, the limiting linear current ramp rate for a successful cold start at $-25\text{ }^{\circ}\text{C}$ is 4 A s^{-1} . If the same current ramp rate is applied at temperatures below $-25\text{ }^{\circ}\text{C}$, failure is expected due to excessive voltage drop in the end-plate cells.

In the $-30\text{ }^{\circ}\text{C}$ cold start test, a linear current ramp rate of 3 A s^{-1} is

applied in Case 13 to avoid startup failure. Compared to Case 12, the startup time of Case 13 increased by approximately 10 s. As shown in Fig. 10(c), the voltage behavior of Cells 1# and 209# in Case 13 is largely consistent with that in Case 12, with both reaching a minimum voltage of 0.18 V during startup. However, the minimum voltages of Cells 2# and 208# are 0.45 V—higher than the 0.38 V observed in Case 12. These results indicate that employing a 3 A s^{-1} current ramp rate effectively mitigated the further influence of the “end-plate effect” on Cells 2# and 208#, significantly improving their internal water-thermal distribution and enhancing the overall cold start performance of the stack.

As shown in Fig. 11(a), an analysis of the coolant temperature rise rate at the end of startup for Cases 10–13 indicates that, compared to the step loading strategy, the linear loading approach enables a more rapid increase in coolant temperature, significantly shortening the cold start duration. At the end of startup, the cell voltage range, illustrated in Fig. 11(b), shows noticeable improvement. Even under $-30\text{ }^{\circ}\text{C}$ conditions, a relatively small voltage range is achieved. Furthermore, the voltage distribution across cells in Fig. 11(c) demonstrates good uniformity in all cases except Case 12, which exhibits poorer voltage balance. These results confirm that the linear loading strategy is more conducive to successful cold starts of the fuel cell stack.

To more clearly illustrate the effect of the linear loading strategy on improving cell voltage uniformity within the stack, the standard deviation of cell voltage C_v at the end of startup is calculated based on Eq. (4). A higher C_v value indicates poorer voltage uniformity among cells. The results are shown in Fig. 12.

As observed, Cases 1 to 6, which achieved successful startup using the step loading strategy, all exhibited C_v values below 0.05. In contrast, Cases 7 to 9, which resulted in startup failure, showed C_v values between 0.05 and 0.10, indicating significantly degraded voltage uniformity and an imbalanced internal water-thermal state within the stack.

The C_v values under the linear loading strategy are markedly lower than those under the step loading approach. Specifically, compared to

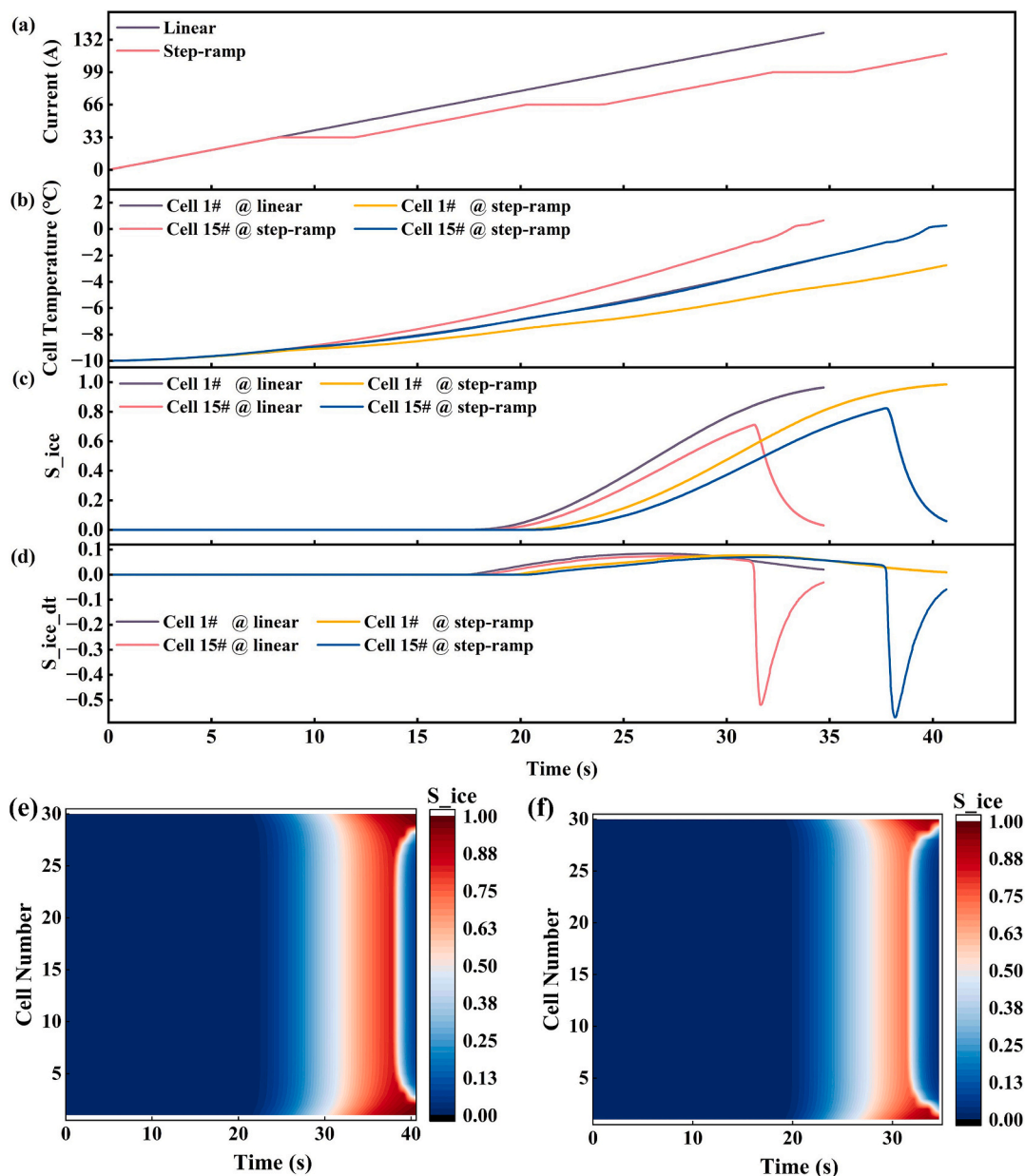


Fig. 9. Simulation analysis of the cold start process: (a) Applied current; (b) Cell temperature; (c) Ice volume fraction; (d) Rate of change of ice volume fraction; (e) Ice volume distribution in the stack under step-loading strategy; (f) Ice volume distribution in the stack under linear-loading strategy.

Table 4

Cold start results with linear loading strategy.

Case	Ambient temperature (°C)	Current loading rate (A/s)	Operating time (s)	Coolant temperature (°C)	Result
10	-15	4	44	0.10	Success
11	-20	4	51	0.41	Success
12	-25	4	50	0.94	Success
13	-30	3	60	0.10	Success

the C_v values of Cases 1, 4, and 8—which are tested at the same temperatures—Cases 10, 11, and 12 achieved reductions of 80.67%, 30.49%, and 4.46%, respectively. These results demonstrate that the linear loading strategy not only enables colder and faster cold starts but also significantly improves cell voltage uniformity within the stack.

4. Conclusions and perspectives

A reliable unassisted cold start capability of fuel cell systems under low-temperature conditions contributes to enhancing their environmental adaptability, reducing system costs, and broadening commercialization prospects. However, current research predominantly focuses on the single-cell level, leaving studies on commercial high-power fuel cell stacks insufficient, particularly regarding the cold-start failure mechanisms of endplate cells under different current loading strategies during startup. To address this gap, this study conducted multiple sets of cold-start experiments at various ambient temperatures, with a focus on analyzing the influence mechanisms of current loading strategies during startup on coolant temperature and individual cell voltage uniformity, especially their impact on the cold-start output performance of endplate cells. Furthermore, a one-dimensional cold-start model was employed to investigate the cold-start behavior of endplate cells, leading to the development of an optimized strategy for mitigating low voltage in these

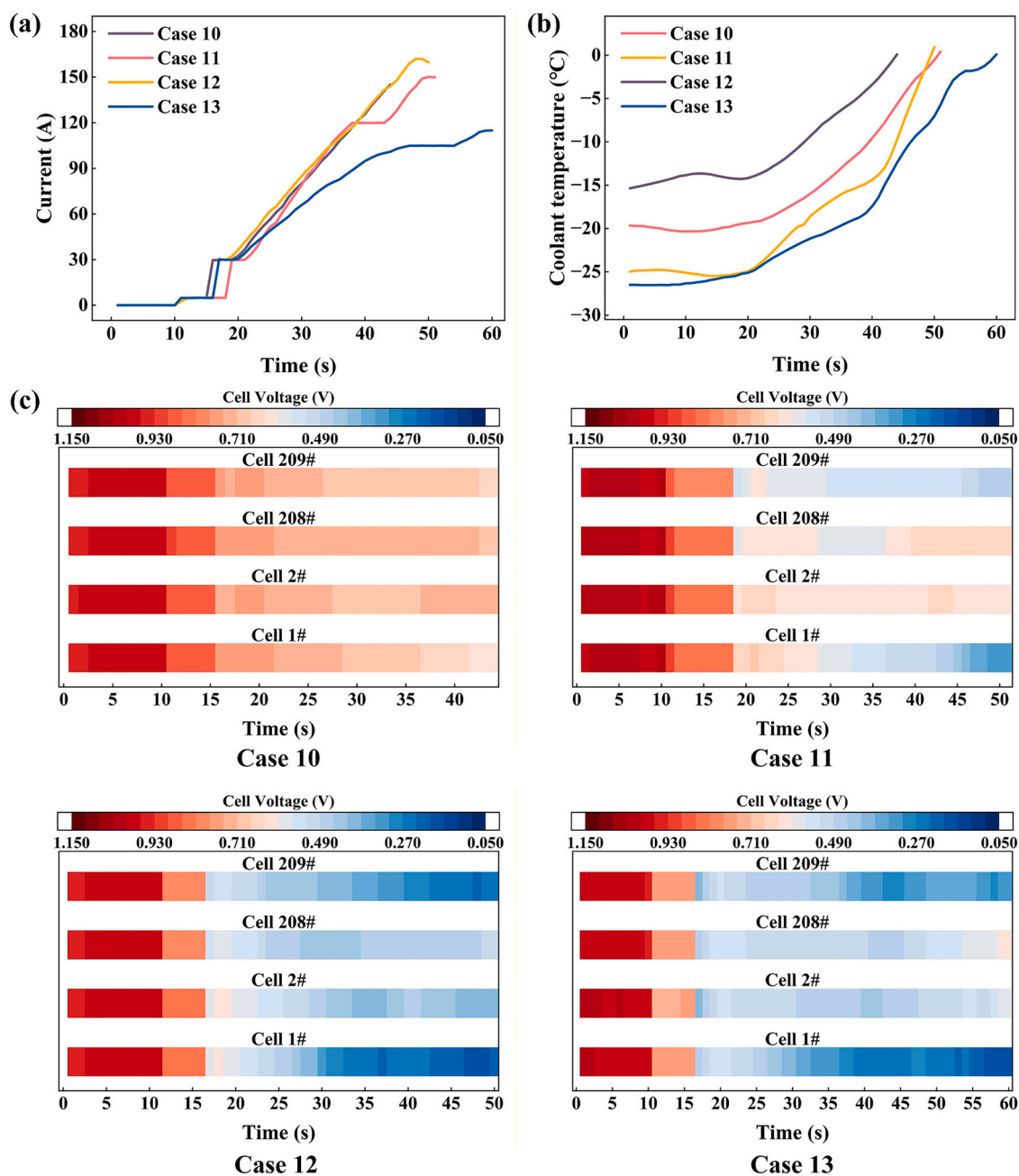


Fig. 10. Cold start with linear loading strategy: (a) applied current; (b) coolant temperature; (c) voltage of end-plate cells.

cells. This strategy was subsequently validated through systematic experiments. The main conclusions are as follows:

- (1) The primary cause of cold-start failure is the sudden performance degradation of endplate cells. Under the influence of the “end-plate effect,” the voltage of endplate cells, particularly Cell #1, exhibits an initial rise followed by a decline during startup. In the later stages of startup, ice formation blocks the reactant gas transport channels, leading to a rapid increase in concentration polarization and a consequent sharp voltage drop, ultimately resulting in startup failure.
- (2) The outcome of cold-start is determined by different current loading strategies through their impact on the water–thermal balance. Simulation analysis reveals that a step loading strategy disrupts the water–thermal balance in end-plate cells, causing a rapid increase in the ice volume fraction within the cell and

consequently leading to startup failure. In contrast, although a linear loading strategy also leads to an increase in ice volume fraction, the concomitant rise in heat generation from the increasing current allows the central cells to gradually gain dominance in the “heat generation vs. ice formation” competition—after the ice volume fraction reaches 0.71, it begins to decline in these cells. The surplus heat can then assist the end-plate cells in achieving a temperature breakthrough before complete ice formation, enabling a successful start.

- (3) The linear loading strategy significantly enhances both the cold-start performance and voltage uniformity of the stack. Both simulation and experimental results demonstrate that compared to the step loading strategy, the linear loading strategy effectively improves the cold-start capability of the fuel cell system, reducing startup time by 7 s and 2 s at $-15\text{ }^{\circ}\text{C}$ and $-20\text{ }^{\circ}\text{C}$, respectively, and achieving successful unassisted starts at $-25\text{ }^{\circ}\text{C}$ and $-30\text{ }^{\circ}\text{C}$.

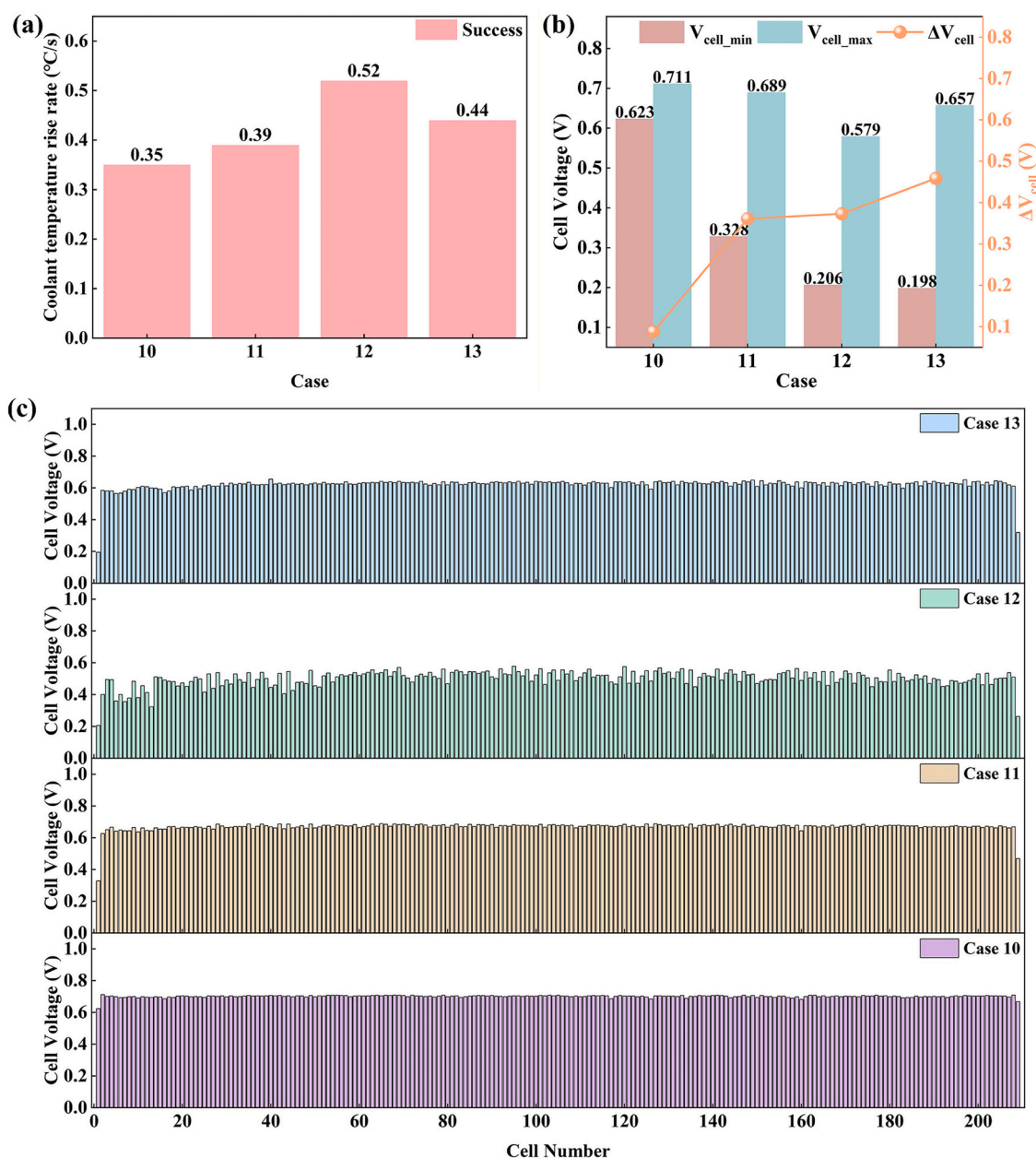


Fig. 11. Analysis of the cold start process under a linear loading strategy: (a) coolant temperature rise; (b) maximum and minimum cell voltages at the end of startup; (c) cell voltage distribution at the end of startup.

Furthermore, the linear loading strategy effectively mitigates the low-voltage issue in end-plate cells, reducing the C_v under the same temperatures by 80.67%, 30.49%, and 4.46%, respectively, thereby significantly improving the overall voltage uniformity of the stack.

Based on these findings, future research should be deepened along the following directions: first, leveraging in-situ characterization and multi-physics simulations to reveal the water-ice phase transition and transport blockage mechanisms at the microscale pore level; second, developing intelligent adaptive control algorithms that integrate linear, pulsed, and other strategies to achieve dynamic optimization; third, pursuing fundamental optimization at both the material level (e.g., high thermal conductivity, ice phobic coatings) and the structural level (e.g., thermal management of endplates); fourth, validating the robustness of these strategies under more extreme environmental conditions and advancing their integration and coordination with the vehicle's thermal

management system.

This research not only holds significant academic value for constructing accurate cold-start models at the stack scale but also carries direct engineering application and industrial significance for addressing the bottlenecks of fuel cell vehicles in cold climates, reducing system costs, and promoting low-carbon transition in the transportation sector.

CRedit authorship contribution statement

Tiancai Ma: Writing – original draft. **Chang Du:** Software. **Ziheng Gu:** Data curation. **Weikang Lin:** Formal analysis. **Yuewen Zeng:** Methodology. **Julong Zhou:** Methodology. **Yahui Yi:** Resources. **Naiyuan Yao:** Project administration. **Liqin Qian:** Validation. **Lei Shi:** Writing – review & editing. **Ruitao Li:** Writing – review & editing.

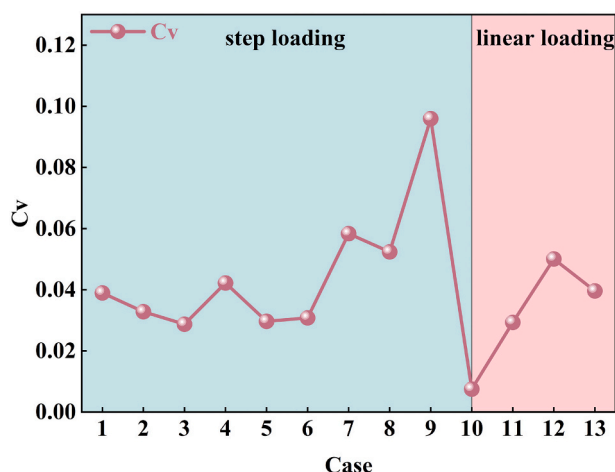


Fig. 12. C_v at the end of the cold start process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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